

Grid Reliability Analytics

Azure SQL + Power BI audit of 526K U.S. outage records that caught an 8.49x overstatement

Role: Data / analytics engineer

T-SQL

Azure SQL

Python

pandas

Power BI

DAX

Built a layered Azure SQL warehouse and Power BI dashboard over 526K U.S. electric outage records (2014-2023), with an explicit data-quality audit layer between source and report. The audit found that a naive sum overstates customer-hours lost by 8.49x.

8.49x**Overstatement Caught**

21.5B naive vs 2.53B customer-hours at the correct grain

526K**Source Rows Audited**

168,858 unique outage spells across 48 states

20**Data-Quality Rules**

273,291 duplicate rows quarantined, not deleted

Problem

Public outage data looks analysis-ready, but exact duplicates and many-to-many event attribution quietly inflate every additive measure. A dashboard built straight on the source would report numbers several times too high.

Solution

The raw data lands as text, so no defect is silently coerced away. A 20-rule audit layer quarantines bad rows instead of deleting them, the model is built at the correct grain, and the report has a data-quality page instead of hiding the caveats.

Architecture

- Python download and raw loader into Azure SQL (all NVARCHAR, nothing rejected or coerced)
- Staging layer typed with TRY_CONVERT only
- dq schema: 20 rules with severity (BLOCK / WARN / INFO) and quarantine tables
- Dimensions, fact table, and a bridge for the many-to-many spell-to-event relationship
- Mart views: the only surface Power BI touches (Import mode)
- Independent pandas expected-results check plus verify.sql

Highlights

- Found that 51.94% of source rows are surplus exact-duplicate copies
- Layered warehouse: raw (all text), staging, data quality, dims/facts, mart views, then Power BI
- A bridge table for the many-to-many spell-to-event relationship stops a second round of double counting
- Rows always reconcile: landed = surviving + quarantined
- Headline KPIs checked against an independent pandas calculation
- External check against EIA-861: $r = 0.793$ across 45 states, with the magnitude gap reported honestly

Challenges and Tradeoffs

- Challenge: SQL Server ignores trailing spaces in comparisons, so a TRIM check found only 16 of 327 whitespace defects. Change: Switched to DATALENGTH.
- Challenge: event_id isn't unique and repeats across years for unrelated events. Change: Keyed events on (event_id, began, restored, cause).
- Challenge: 75 raw cause strings covering typos, mixed conventions, and NBSP delimiters. Change: Mapped them to 10 categories with documented precedence.
- Tradeoff: A third drill-through page vs verified numbers. Decision: Cut the page and shipped two pages that reconcile to the independent check.

Outcomes

- Recovered the correct customer-hours total (2.53B vs a naive 21.5B)
- The data-quality page lists all 20 rules, including the 14 that passed
- Limitations documented openly: no true SAIDI/SAIFI, trend not coverage-adjusted, results depend on attribution

Repository: <https://github.com/Jaskeerat/grid-reliability-analytics> · Full case study: <https://jaskeerat-portfolio.vercel.app/projects/grid-reliability-analytics>